Solar E-Rickshaw



Category: **Solar, Power & Energy** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Solar e-rickshaw

What it is used for

A solar e-rickshaw is an electric three-wheeler that may use solar panels to supplement battery charging or power auxiliary systems. Operate the vehicle according to its manufacturer instructions and treat the traction battery and electrical drive system with appropriate precautions.

Main parts

- Vehicle chassis, wheels, steering, and brakes
- Traction battery and battery management system
- Electric motor and motor controller
- Solar panels and solar charge controller, if fitted
- Charging port, cables, and electrical protection
- Dashboard, lights, and driver controls

Before you start

1. Inspect tyres, brakes, steering, lights, mirrors, and chassis for damage.
2. Check the traction battery status, dashboard warnings, and charging connections.
3. Inspect solar panels, mounting, cables, and controller for visible damage.
4. Confirm the battery charge level is adequate for the planned journey.
5. Check that the passenger and cargo load stays within the rated capacity.
6. Ensure the vehicle is in a safe condition and required documents and protective equipment are available.

How to use

1. Park the vehicle on level ground and apply the parking brake before checks.
2. Switch off the vehicle before inspecting electrical connectors or charging equipment.
3. Confirm that solar and mains charging connections, if fitted, are secure and undamaged.
4. Charge the battery only through the approved charging system and within its rated limits.
5. Check the dashboard for warnings before starting the vehicle.
6. Adjust mirrors and ensure passengers or cargo are secured correctly.
7. Release the parking brake and start moving gradually using the approved controls.
8. Drive at a safe speed appropriate to road conditions, load, and vehicle limits.
9. Monitor battery status, warning lights, steering, brakes, and unusual sounds during travel.
10. Stop safely if a critical warning, loss of power, or electrical fault occurs.
11. Park in a safe location, apply the parking brake, and switch off the vehicle.
12. Disconnect charging equipment only after the charging session has stopped.

3. Record defects and arrange qualified inspection before further use when safety is uncertain.

⚠ Safety

- Only trained and authorised drivers should operate the e-rickshaw.
- Follow local traffic rules, passenger limits, and the manufacturer's rated load.
- Do not drive with faulty brakes, steering, tyres, or critical warning indicators.
- Do not touch exposed high-voltage or traction-battery conductors; repairs require qualified personnel.
- Do not charge a damaged, swollen, or overheated battery.
- Keep the battery and charging area away from fire, water ingress, and unauthorised access.

Care & storage

- Inspect tyres, brakes, steering, lights, and suspension at recommended intervals.
- Keep solar panels clean using the approved method and safe procedure.
- Check battery condition, connectors, and charging equipment according to the maintenance schedule.
- Keep the vehicle dry where required and protect electrical connectors from corrosion.
- Arrange servicing for unusual battery heating, reduced range, or repeated electrical faults.

Common mistakes to avoid

- Exceeding the rated passenger or cargo capacity.
- Ignoring battery warnings or signs of electrical damage.
- Assuming solar panels replace the need for correct battery charging and maintenance.
- Attempting traction-battery repairs without appropriate qualifications.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.